

Luke Bhan

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RESEARCH Machine learning, Neural operators, Learning-based control.

CURRENT **University of California, San Diego**, La Jolla, California Sept 2022 - Present
EDUCATION *Ph.D. Student, Department of Electrical and Computer Engineering*
 Advisors: Yuanyuan Shi, Miroslav Krstic

PREVIOUS **Vanderbilt University**, Nashville, TN Sept 2020 - May 2022
EDUCATION *Master of Science, Department of Computer Science*
 Advisor: Gautam Biswas
 Thesis: Deep Reinforcement Learning for Adaptive Control in Robotics

Vanderbilt University, Nashville, TN Sept 2018 - May 2020
 Bachelor of Science, Department of Computer Science and
 Department of Physics and Astronomy
 Advisors: Gautam Biswas and Kálmán Varga

HONORS AND **Best Paper Award Finalist**, Learning for Dynamics and Control Confer-
AWARDS ence 2025 (3/119. For the work [C10] below)

Commitment to Diversity Award, University of California, San Diego,
 Department of Electrical and Computer Engineering (10k funding)

Underwood Memorial Award, Vanderbilt University, Department of
 Physics and Astronomy, (Outstanding graduating senior in physics)

Best Undergraduate Paper Award, Vanderbilt University, Department
 of Physics and Astronomy, (For the work [J1] below)

GRANTS AND **Computational Science Graduate Fellowship**, Department of Energy,
FELLOWSHIPS USA (4 year Ph.D. funding, 500k)

NVIDIA Academic Grant, Co-Investigator with Yuanyuan Shi, (8x RTX
 PRO 6000 Server Edition GPUs to support *PDEControlGym: Physical AI*
 Benchmarks for RL Control of PDE Systems)

Ph.D. Fellowship, University of California, San Diego, Department of
 Electrical and Computer Engineering (66k funding)

JOURNAL [J10] Luke Bhan, Miroslav Krstic, Yuanyuan Shi. "Stabilization of nonlinear
PUBLICATIONS systems with unknown delays via delay-adaptive neural operator approxi-
 mate predictors." *Automatica, provisionally accepted, 2026*.

 [J9] Luke Bhan, Peter Quawas, Miroslav Krstic, Yuanyuan Shi. "Sampling-
 Horizon Neural Operator Predictors for Nonlinear Control Under Delayed
 Inputs." *IEEE Control Systems Letters, Volume 10, 2026*.

 [J8] Maxence Lamarque, Luke Bhan, Yuanyuan Shi, Miroslav Krstic. "Adap-

tive Neural-Operator Backstepping Control of a Benchmark Hyperbolic PDE.” *Automatica*, Volume 177, 2025.

[J7] Maxence Lamarque, Luke Bhan, Rafael Vazquez, Miroslav Krstic. ”Gain Scheduling with a Neural Operator for a Transport PDE with Nonlinear Recirculation.” *IEEE Transactions on Automatic Control*, Volume 70. 2025.

[J6] Luke Bhan, Yuanyuan Shi, Miroslav Krstic. ”Adaptive control of reaction–diffusion PDEs via neural operator-approximated gain kernels.” *System & Control Letters*, Volume 195. 2024.

[J5] Miroslav Krstic, Luke Bhan, Yuanyuan Shi. ”Neural operators of backstepping controller and observer gain functions for reaction–diffusion PDEs.” *Automatica*, Volume 164. 2024.

[J4] Luke Bhan, Yuanyuan Shi, Miroslav Krstic. ”Neural operators for bypassing gain and control computations in PDE backstepping.” *IEEE Transactions on Automatic Control*, Volume 69. 2023.

[J3] Luke Bhan, Cody L Covington, Kálmán Varga. ”Laser-Driven Peta-hertz Electron Ratchet Nanobubbles.” *Nano Letters*, Volume 22. 2022.

[J2] Luke Bhan, Cody L Covington, Kálmán Varga. ”Signatures of atomic structure in subfemtosecond laser-driven electron dynamics in nanogaps.” *Physical Review B*, Volume 105. 2022.

[J1] Luke Bhan, Cody L Covington, Jason Rivas, Kálmán Varga. ”Simulation of photo-electron spectrum and electron scattering by dual time propagation.” *The Journal of Chemical Physics*, Volume 154. 2021.

CONFERENCE
PUBLICATIONS

[C12] Filip Bajraktari, Luke Bhan, Miroslav Krstic, Yuanyuan Shi. ”Delay compensation of multi-input distinct delay nonlinear systems via neural operators.” In Proceedings of *American Control Conference (ACC)*, 2026.

[C11] Luke Bhan, Miroslav Krstic, Yuanyuan Shi. ”Delay-adaptive Control of Nonlinear Systems with Approximate Neural Operator Predictors.” In Proceedings of *IEEE Conference on Decision and Control (CDC)*, 2025.

[C10] Luke Bhan*, Peijia Qin*, Miroslav Krstic, Yuanyuan Shi. ”Neural Operators for Predictor Feedback Control of Nonlinear Delay Systems.” In Proceedings of *Learning for Dynamics and Control (L4DC)*, 2025. **Best Paper Finalist.**

[C9] Sharath Matada, Luke Bhan*, Yuanyuan Shi, Nikolay Atanasov. ”Generalizable Motion Planning via Operator Learning.” In Proceedings of *International Conference on Learning Representations (ICLR)*, 2025.

[C8] Luke Bhan*, Yuexin Bian*, Miroslav Krstic, Yuanyuan Shi. ”PDE Control Gym: A Benchmark for Data-Driven Boundary Control of Partial Differential Equations.” In Proceedings of *Learning for Dynamics and Control (L4DC)*, 2024.

[C7] Luke Bhan, Yuanyuan Shi, Iasson Karafyllis, Miroslav Krstic, James B Rawlings. ”Moving-Horizon Estimators for Hyperbolic and Parabolic PDEs in 1-D.” In Proceedings of *American Control Conference (ACC)*, 2024.

[C6] [Luke Bhan](#), Yuanyuan Shi, Miroslav Krstic. "Neural Operators for Hyperbolic PDE Backstepping Feedback Laws." In Proceedings of *IEEE Conference on Decision and Control (CDC)*, 2023.

[C5] [Luke Bhan](#), Yuanyuan Shi, Miroslav Krstic. "Neural Operators for Hyperbolic PDE Backstepping Kernels." In Proceedings of *IEEE Conference on Decision and Control (CDC)*, 2023.

[C4] [Luke Bhan](#), Yuanyuan Shi, Miroslav Krstic. "Operator learning for nonlinear adaptive control." In Proceedings of *Learning for Dynamics and Control (L4DC)*, 2023.

[C3] [Luke Bhan](#), Marcos Quinones-Grueiro, Gautam Biswas. "Concurrent policy blending and system identification for generalized assistive control." In Proceedings of *International Conference on Robotics and Automation (ICRA)*, 2022.

[C2] [Luke Bhan](#), Marcos Quinones-Grueiro, Gautam Biswas. "Fault tolerant control combining reinforcement learning and model-based control." In Proceedings of *International Conference on Control and Fault-Tolerant Systems (SysTol)*, 2021.

[C1] Adam Stager, [Luke Bhan](#), Andreas Malikopoulos, Liuhui Zhao. "A Scaled Smart City for Experimental Validation of Connected and Automated Vehicles." In Proceedings of *IFAC Symposium on Control in Transportation Systems (CTS)*, 2018.

WORKSHOP PAPERS

[W1] [Luke Bhan](#), Hanyu Zhang, Andrew Gordon Wilson, Michael W. Mahoney, Chuck Arvin. "The Forecast Critic: Leveraging Large Language Models for Poor Forecast Identification." *AAAI AI4TS and AABA4ET Workshops*, 2026.

PREPRINTS

Miroslav Krstic, [Luke Bhan](#). "Families of Control-Cost-Parametrized Inverse-Optimal Universal Stabilizers." Submitted to *System & Control Letters*.

Miroslav Krstic, Iasson Karafyllis, [Luke Bhan](#), Carina Veil. "Lotka-Sharpe Neural Operators for Control of Population PDEs." Submitted to *IEEE Transactions on Automatic Control (TAC)*.

[Luke Bhan](#), Miroslav Krstic, Yuanyuan Shi. "Predictor-Based Output-Feedback Control of Linear Systems with Time-Varying Input and Measurement Delays via Neural-Approximated Prediction Horizons." Submitted to *System & Control Letters*.

[Luke Bhan](#), Miroslav Krstic, Yuanyuan Shi. "Neural operators for predictor feedback control of nonlinear delay systems." Submitted to *IEEE Transactions on Automatic Control (TAC)*.

INTERNSHIPS

Research Scientist Intern, Microsoft June 2026 - Current
On-going. Developing foundation models for power systems (AC-OPF).

Applied Scientist Intern, Amazon June 2025 - Sept 2025
Fine-tuned and designed custom LLMs to provide adjustments for time-series forecasts predicting demands of over 30+ million Amazon products. Demonstrated that LLMs can effectively identify poor distributional forecasts which led to an internal Amazon Machine Learning Conference (AMLC)

paper.

Research Intern, *Lawrence Berkeley National Lab* June 2024 - Aug 2024
Developed randomized linear algebra (RandLA) solvers for distributed non-convex optimization (SQP) enabling real-time, parallel optimization for control of dynamical systems.

Software Engineering Intern, *MongoDB* June 2021 - Aug 2021
Created the compression algorithm for MongoDB's time-series database.

Machine Learning Intern, *T-Mobile* June 2020 - Aug 2020
Created an internal analytics API for visualizing network loads to proactively identify and combat downtime.

TEACHING

Co-Instructor Physics Informed Machine Learning, University of California San Diego. Spring 2026, Spring 2025

- Taught 5 Lectures on Foundation Models, Neural ODEs, Neural Operators, and Physics Informed Neural Networks (PINNs).
- Created and developed custom homework from scratch exploring augmented neural ODEs, FNOs, and PINNs.

Teaching Assistant *Numerical Analysis*, Vanderbilt University Mathematics Department Fall 2021

Teaching Assistant *Introduction to Probability and Statistics*, Vanderbilt University Mathematics Department Fall 2021

Teaching Assistant, *Intermediate Software Design* Vanderbilt University Computer Science Department Fall 2020 - Fall 2021

INVITED TALKS

Seminar Presentations

Seminar, University of Washington (Hosted by Prof. Baosen Zhang) March 2026

Seminar, University of California, San Diego (Hosted by Prof. Sylvia Herbert) Sept. 2025

Fed-NeMO Seminar, Yale University, (Hosted by Prof. Lu Lu) Oct. 2024

Seminar, International Institute of Computer Science, UC Berkeley (Michael Mahoney's Group) June 2024

Seminar, Hong Kong University of Science and Technology (Huan Yu's Group) March 2024

Internal Control Seminar, University of California, San Diego (Hosted by Yang Zheng) April 2023

Conference Presentations

Paper presentation, American Control Conference (ACC) July 2024

Paper presentation, IEEE Conference on Decision and Control (CDC) Dec. 2024

Poster presentation, Learning for Dynamics and Control (L4DC) June 2023

Paper presentation, International Conference on Robotics and Automation (ICRA) June 2022

Paper presentation, International Conference on Control and Fault-Tolerant Systems (SysTol) Feb. 2022

ACADEMIC SERVICE

Workshop Organizer

PDE-AI Workshop: AI for Modeling, Control, and Optimization of Partial Differential Equations, *American Control Conference (ACC)* (2026).
acc-2026-ai-for-pde-workshop.vercel.app

Journal Reviewer

Automatica (2023-)

IEEE Transactions on Automatic Control (2023-)
Systems & Control Letters (2024-)
International Journal of Robust and Nonlinear Control (2024-)
IEEE Control Systems Letters (2025-)
IEEE Transactions on Control of Networked Systems (2026-)

Conference Reviewer

IEEE Conference on Decision and Control (CDC), (2023-)
American Control Conference (ACC), (2024-)
Learning for Dynamics and Control (L4DC), (2022-)
Association for the Advancement of Artificial Intelligence Conference (AAAI),
(2023-)
International Conference on Learning Representations (ICLR), (2023-)